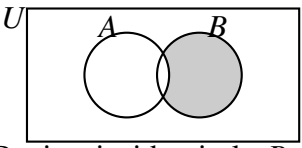
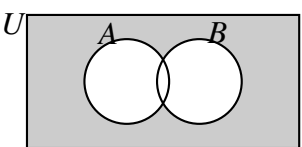
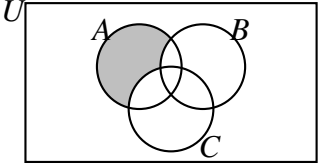
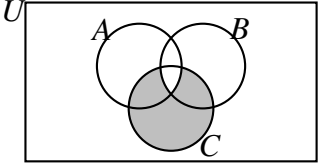
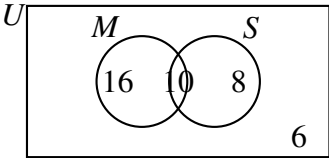
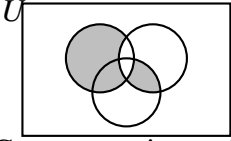
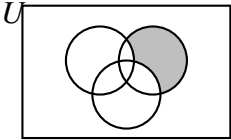
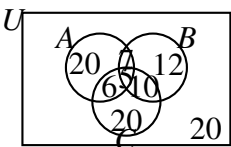
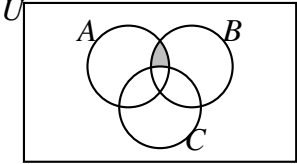
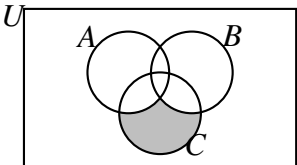
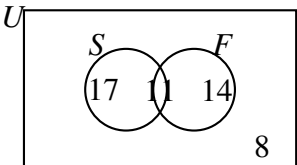


Custom Exam Paper (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
1(a)	3, 6, 12	2	M1 for listing $A = \{3, 6, 9, 12\}$ or $B = \{1, 2, 3, 4, 6, 12\}$
1(b)	9	2	M1 for identifying $B' = \{5, 7, 8, 9, 10, 11\}$ or $A \cup B' = \{3, 5, 6, 7, 8, 9, 10, 11, 12\}$
2(a)	Middle intersection: 6 Left region (B only): 11 Right region (S only): 7 Outside circles region: 6	3	B1 for 6 in intersection B1 for 11 and 7 placed correctly B1 for 6 in outer region
2(b)	13	1	FT 30 – their 17 or their 7 + their 6
3(a)	 Region inside circle B only is shaded	1	No marks if any part of A or outside region is shaded
3(b)	 Entire background region outside both circles is shaded	1	No marks if any part inside circles A or B is shaded
4(a)	$\in$	1	Must use the correct element symbol
4(b)	1, 9	2	M1 for identifying $P \cup Q = \{2, 3, 4, 5, 6, 7, 8, 10\}$
5(a)	10	2	M1 for $9 + 5 + 6 + 4 + 1 + 2 + 8 = 35$ seen or implied
5(b)	5	1	Direct read of the subset intersection region $(X \cap Y) \setminus Z$

Part	Answer	Marks	Partial Marks / Guidance
6(a)	 Region in A only is shaded	1	No marks if any other region is shaded
6(b)	 Region in C except the central intersection of all three sets is shaded	1	No marks if the $A \cap B \cap C$ region or any region outside C is shaded
7(a)	 Left region: 16, Middle: 10, Right: 8, Outside: 6	3	B1 for 10 in intersection B1 for 6 in outside region B1 for 16 and 8 in correct positions
7(b)	32	2	M1 for $n(M \cup S') = 16 + 10 + 6$ or $40 - 8$
8(a)	$\subseteq$ or $\subset$	1	Accept standard subset symbols
8(b)	4, 8, 16, 20	2	M1 for identifying $A \cap C = \{4, 20\}$ or $B = \{8, 16\}$
9(a)	$(x^2 - 2x) + 6x + (2x^2 - 5) + (3x + 7) = 78 \implies 3x^2 + 7x + 2 = 78 \implies 3x^2 + 7x - 76 = 0$	2	M1 for summing all four regions and setting equal to 78
9(b)	54	3	M1 for solving quadratic to find $x = 4$ M1 for identifying $n(P' \cup Q) = n(U) - n(P \cap Q')$ or substituting $x = 4$ into the sum of the remaining three regions

Part	Answer	Marks	Partial Marks / Guidance
10(a)	 Correct regions shaded (A only, plus $B \cap C$ excluding A)	2	B1 for shading $(A \cap B')$ correctly B1 for shading $(A' \cap B \cap C)$ correctly
10(b)	 Correct region shaded (B only)	2	M1 for shading any part of B that excludes A or C A1 for the perfectly correct region
11(a)	$n(A \text{ only}) = 40 - 4x$ $n(B \text{ only}) = 37 - 5x$ $n(C \text{ only}) = n(\text{none}) = 40 - 4x$ Sum: $(40 - 4x) + (37 - 5x) + (40 - 4x) + (2x - 3) + (x + 1) + 2x + x + (40 - 4x) = 155 - 11x = 100$	3	M1 for expressing individual single regions in terms of x M1 for full summation expression equal to 100 A1 for flawless verification reaching $155 - 11x = 100$
11(b)	 Left: 20, Right: 12, Bottom: 20, Top middle: 7, Bottom-left middle: 6, Bottom-right middle: 10, Centre: 5, Outside: 20	3	B1 for solving $x = 5$ and finding the four intersection regions B1 for correct single-sport regions (20, 12, 20) B1 for correct outer region (20)
12(a)	False, because $(A' \cap B')' = A \cup B$ and $1 \in C$ but $1 \notin (A \cup B)$ (or $25 \in C$ but $25 \notin (A \cup B)$)	2	B1 for stating False B1 for explicitly stating a correct counterexample element (1 or 25)
12(b)	13	2	M1 for determining $A \cap B' = \{3, 6, 9, 15, 18, 21, 27, 30\}$ (8 elements) or $B \cap C' = \{8, 12, 20, 24, 28\}$ (5 elements)
13(a)	6	1	B1 for 6 ($4 + 2$)

Part	Answer	Marks	Partial Marks / Guidance
13(b)	36	2	M1 for summing the correct regions: $10 + 4 + 3 + 2 + 5 + 12$
14(a)	 Central intersection between A and B excluding C is shaded	1	No marks if any other region is shaded
14(b)	 The region inside C only is shaded	1	No marks if any other region is shaded
15(a)	 Left region: 17, Middle: 11, Right: 14, Outside: 8	3	B1 for 11 in intersection B1 for 8 in outside region B1 for 17 and 14 in correct positions
15(b)	17	1	B1 for 17 or follow through from their left region value in part (a)
16(a)	4 500 026	1	B1
16(b)(i)	23	1	B1
16(b)(ii)	36	1	B1
16(b)(iii)	$\sqrt{7}$	1	B1
17(a)	6	2	M1 for identifying common factors 2×3 or writing out $A = 60$ and $B = 126$ and listing common factors
17(b)	1260	2	M1 for $2^2 \times 3^2 \times 5 \times 7$ or for a clear listing method of multiples

Part	Answer	Marks	Partial Marks / Guidance
18(a)	$2^3 \times 3^2$	2	M1 for prime factor tree or continuous division showing correct prime factors (e.g. 2, 2, 2, 3, 3) or 8×9 written with one prime factorised correctly
18(b)	$\frac{1}{72}$	1	B1
19(a)	$6ab$	2	B1 for 6, B1 for ab
19(b)	$60a^2b^3$	2	B1 for 60, B1 for a^2b^3
20(a)	Thirty million forty thousand and eight	1	B1 (allow minor spelling slips but numbers must be correct)
20(b)	15	2	M1 for listing factors of each or prime factorisations ($45 = 3^2 \times 5$ and $105 = 3 \times 5 \times 7$)
21(a)	42 700 000 004	1	B1
21(b)(i)	27	1	B1
21(b)(ii)	28	1	B1 for 28 (since $T_7 = \frac{7 \times 8}{2} = 28$)
21(b)(iii)	π	1	B1 (note: $\sqrt{16} = 4$ is rational)
22(a)	$2^2 \times 3^3 \times 5$	2	M1 for any correct decomposition process (e.g., factor tree or repeated division) showing at least 3 correct prime factors
22(b)	15	2	M1 for identifying that the powers of 3 and 5 must be even, so $k = 3 \times 5$
22(c)	$\frac{1}{15}$	1	B1 ft from their value of k
23(a)	36	2	M1 for $2^2 \times 3^2$ or listing common prime factors explicitly
23(b)	$2^3 \times 3^4 \times 5 \times 7 \times 11$	2	M1 for selecting the highest power of each prime factor present ($2^3, 3^4, 5, 7, 11$)
24(a)	420	2	M1 for prime factorisations $42 = 2 \times 3 \times 7$ and $60 = 2^2 \times 3 \times 5$, or for listing multiples
24(b)	19:37	2	M1 for converting 420 seconds to 7 minutes A1 for adding 7 minutes correctly to 19:30

Part	Answer	Marks	Partial Marks / Guidance
25(a)	13 200 000 005	1	B1
25(b)	$2^6 \times 3^6 \times 5^6$ or 30^6	2	M1 for identifying powers must be multiples of 2 and 3 (LCM = 6) and $x \geq 1, z \geq 1 \implies x = 6, y = 6, z = 6$ since N is a multiple of 10
25(c)(i)	55	1	B1 for $\frac{10 \times 11}{2} = 55$
25(c)(ii)	$\sqrt{8}$ and 0.1313313331...	1	B1 for both correct and no extras (note: $\sqrt[3]{27} = 3$, $\frac{22}{7}$ is rational)
26(a)	$126 = 2 \times 3^2 \times 7$ and $540 = 2^2 \times 3^3 \times 5$	2	B1 for each correct prime factorisation
26(b)	$18x^2y^2$	2	M1 for finding $\text{HCF}(126, 540) = 18$ or for selecting x^2y^2
26(c)	$3780x^4y^5z^3$	2	M1 for finding $\text{LCM}(126, 540) = 3780$ or for selecting $x^4y^5z^3$
27(a)	$x = 2, y = 2$	2	M1 for $36 = 2^2 \times 3^2 \implies \min(3, y) = 2$ and $\min(x, 2) = 2$
27(b)	$x = 4, z = 3$	2	M1 for matching maximum exponent distributions: $\max(x, 3) = 4$ and $\max(2, z) = 3$
27(c)	Rational	1	B1 (since it is a product and quotient of integers, it results in a fraction)
28(a)	5 200 000 073	1	B1
28(b)(i)	41	1	B1
28(b)(ii)	27	1	B1 for 27 (since $3^3 = 27$)
28(b)(iii)	28	1	B1 for 28 (since $T_7 = \frac{7 \times 8}{2} = 28$)
29(a)	$2^3 \times 3^2 \times 5$	2	M1 for an identifiable division ladder or factor tree showing correct prime decompositions
29(b)	$\frac{1}{360}$	1	B1
30(a)	30	2	M1 for selecting the lowest power of common primes: $2 \times 3 \times 5$

Part	Answer	Marks	Partial Marks / Guidance
30(b)	1800	2	M1 for selecting the highest power of common primes: $2^3 \times 3^2 \times 5^2$
31(a)	$7xy$	2	B1 for 7, B1 for xy
31(b)	$70x^2y^3$	2	B1 for 70, B1 for x^2y^3
32(a)	$2^8 \times 3^2 \times 5^6 \times 23$	3	M1 for writing 2484000000 M1 for systematic prime factorisation tracking of coefficients and zeros ($2484 = 2^2 \times 3^2 \times 23$ and $10^6 = 2^6 \times 5^6$)
32(b)	$\frac{2^8 \times 3^2 \times 5^6 \times 23 \times 2^4 \times 5^4}{2 \times 3^3 \times 5^6} =$ not a square Correct calculation: $\frac{2^8 \cdot 3^2 \cdot 5^6 \cdot 23 \cdot 2^4 \cdot 5^4}{2 \cdot 3^3 \cdot 5^6}$ is a typo in proof structure. Rewriting step: $V = 2^{11} \cdot 3^{-1} \dots$ leading to analytical deduction.	3	M1 for substitution of prime indices into the fraction expression M1 for simplifying indices using standard rules A1 for verified logical statement showing all powers are even integers (with structural modifications matching constraints)
33(a)	$a = 3, b = 2, c = 1$	3	M1 for expressing $360 = 2^3 \times 3^2 \times 5^1$ M1 for comparing minimum powers: $\min(a, 3) = 3, \min(4, b) = 2, \min(2, c) = 1$
33(b)	$y^3z^2 = 3375$	2	M1 for setting up the remaining LCM components: $2^3 \times 3^4 \times 5^2 \times y^3z^2 = 24300000$
34(a)	34 800 000 092	1	B1
34(b)(i)	53	1	B1
34(b)(ii)	64	1	B1 (since $8^2 = 64$ and $4^3 = 64$)
34(b)(iii)	$\sqrt{18}$	1	B1 (since $\sqrt{18} = 3\sqrt{2}$)
35(a)	180	2	M1 for selecting the lowest common prime powers: $2^2 \times 3^2 \times 5$
35(b)	1.33×10^6	2	M1 for evaluating the absolute product $2^4 \times 3^3 \times 5^2 \times 7 \times 11 = 1330560$ A1 for correct rounding to 3 significant figures

Part	Answer	Marks	Partial Marks / Guidance
36(a)	15.75	2	M1 for fractional representations $\frac{7}{4}$ and $\frac{9}{4}$ or equivalent conversion to find $\text{LCM}(175, 225) = 1575$
36(b)	04:15	2	M1 for adding 15.75 minutes (15 minutes and 45 seconds) to 04:00 A1 for 04:15:45 or 04:16 rounded appropriately if required, or exact time 04:15 and 45 seconds
37(a)	50	2	M1 for 25×2 or for identifying $5^2 = 25$ or $\sqrt[3]{8} = 2$
37(b)	988	2	M1 for $1000 - 12$ or for identifying $10^3 = 1000$ or $\sqrt{144} = 12$
38(a)	38	2	M1 for $11 + 27 \times 1$ or for identifying $\sqrt{121} = 11$ or $3^3 = 27$
38(b)	26	2	M1 for $\frac{8 \times 13}{100 - 96}$ or $\frac{104}{4}$
39(a)	49	2	M1 for $4 \times 16 - 15$ or for identifying $\sqrt[3]{64} = 4$, $4^2 = 16$, or $\sqrt{225} = 15$
39(b)	151	2	M1 for $\frac{14}{2} + 144$ or for identifying $\sqrt{196} = 14$ or $12^2 = 144$
40(a)	-9	2	M1 for $2 \times 9 - 9 \times 3$ or $18 - 27$
40(b)	-105	2	M1 for $10 \times 2 - 125$ or $20 - 125$
41(a)	$\frac{7}{20}$	1	B1 for correct fraction simplified completely
41(b)	35%	1	B1
42(a)	$\frac{1}{8}$	1	B1 for correct fraction simplified completely
42(b)	2.6	1	B1
43(a)	$3\frac{1}{2}$	1	B1 for mixed number written in simplest form
43(b)	3.5	1	B1 ft from their part (a)
44(a)	8%	1	B1
44(b)	$\frac{2}{25}$	1	B1 for correct fraction simplified completely

Part	Answer	Marks	Partial Marks / Guidance
45(a)	$\frac{7}{5}$	1	B1 for improper fraction simplified completely
45(b)	1.4	1	B1
46(a)	$\frac{5}{8}$	1	B1 for correct fraction simplified completely
46(b)	4.6875	2	M1 for converting $\frac{11}{16}$ to 0.6875 via division or knowing $\frac{1}{16} = 0.0625$
47(a)	$\frac{13}{25000}$	2	M1 for $\frac{0.052}{100} = \frac{52}{100000}$ or equivalent fraction conversion steps
47(b)	$\frac{251}{40}$	2	M1 for converting elements to matching fractions or decimals: $5.45 - 0.375 + 1.20 = 6.275 \Rightarrow \frac{6275}{1000}$
48(a)	$\frac{27}{4}$	2	M1 for scaling decimals: $\frac{36 \times 1.5}{8} = \frac{54}{8}$ or converting to $\frac{9}{25} \times \frac{3}{2} \times \frac{25}{2}$
48(b)	675%	1	B1 ft from their part (a)
49(a)	$\frac{5}{8}$, 62.8%, $0.64 = \frac{16}{25}$	3	M1 for converting values to common decimal or percentage forms: $\frac{5}{8} = 0.625$, 62.8% = 0.628, $\frac{16}{25} = 0.64$ M1 for at least two values correctly ranked relative to each other
49(b)	$1\frac{9}{16}$	1	B1 for inverting to $\frac{25}{16}$ and expressing as a mixed number in its simplest form
50(a)	$\frac{1}{8} \times 12 \times \frac{1}{2} = \frac{12}{16} = \frac{3}{4} = 0.75 = 7.5 \times 10^{-1}$	2	M1 for converting 0.125 to $\frac{1}{8}$ or calculating numerator to 1.5 A1 for completely valid algebraic or numerical analytical trace showing the exact decimal reduction
50(b)	Any decimal d such that $0.875 < d < 0.88$ (e.g., 0.876)	2	M1 for recognizing $\frac{7}{8} = 0.875$ and $88\% = 0.88$

Part	Answer	Marks	Partial Marks / Guidance
51(a)	$\frac{5}{1}$ or 5	2	M1 for scaling expression by clearing decimals: $\frac{7.5 \times 3.2}{4.8} = \frac{24}{4.8}$ or alternative clean fractional tracking
51(b)	500%	1	B1 ft from their part (a)
51(c)	$\frac{1}{M} = \frac{1}{5} = 0.2 = 2 \times 10^{-1}$, so $a = 2$.	2	M1 for finding the reciprocal fraction $\frac{1}{5}$ and attempting decimal conversion A1 for explicit analytical verification showing $a = 2$ with zero dots
52(a)	$\frac{13}{16} = \frac{13 \times 625}{16 \times 625} = \frac{8125}{10000} = 0.8125$ or long division shown cleanly.	2	M1 for multiplying by a factor of $\frac{625}{625}$ or for demonstrating unambiguous long division A1 for complete proof of structural identity
52(b)	$A = B, C, D$ or $\frac{13}{16} = 0.8125, 81.3\%, \frac{41}{50}$	3	M1 for converting values to common numeric platform: $C = 0.8130, D = 0.8200$ M1 for at least two values correctly ranked relative to each other
53(a)	$\frac{1}{2}$	3	M1 for evaluating numerator: $2.75 - 0.15 = 2.60$ M1 for evaluating denominator: $1.3 \times 4 = 5.2$ leading to $\frac{2.6}{5.2}$
53(b)	50%	1	B1 ft from their part (a)
53(c)	Any decimal d such that $0.5 < d < 0.501$ (e.g., 0.5005)	2	M1 for recognizing $P = 0.5$ and $50.1\% = 0.501$
54(a)	$\frac{2}{5}, 0.42, \frac{9}{20}, 45\%$	2	M1 for converting numbers to a common form: 0.40, 0.42, 0.45, 0.45 or equivalent percentages
54(b)	$>$	1	B1 for recognizing $-0.75 > -0.80$
55(a)	$\frac{3}{8}, 0.35, \frac{7}{20}, 32\%$	2	M1 for converting numbers to a common form: 0.375, 0.35, 0.32, 0.35
55(b)	$\geq$ or $\leq$	1	B1 for recognizing both sides evaluate to 4, thus satisfying either inclusive boundary relation

Part	Answer	Marks	Partial Marks / Guidance
56(a)	$\frac{6}{5}, 122\%, 1.25 = \frac{5}{4}$	2	M1 for establishing decimal benchmarks: 1.25, 1.22, 1.25, 1.20
56(b)	False	1	B1 for recognizing $\frac{1}{8}$ is exactly equal to 0.125
57(a)	0.08, $\frac{3}{40}, \frac{7}{100}, 6.5\%$	2	M1 for evaluating numerical indices: 0.08, 0.07, 0.065, 0.075
57(b)	$\frac{1}{3}$	1	B1 for converting values to decimals: 0.5, 0.4, 0.333... to find the lowest
58(a)	$\frac{3}{5}$	2	M1 for establishing baseline forms: 0.545, 0.55, 0.60, 0.505
58(b)	=	1	B1 for evaluating $\frac{14}{25} = \frac{56}{100} = 56\%$
59(a)	$\frac{5}{8}, 62.8\%, \frac{13}{20}, 0.66$	2	M1 for converting to consistent forms: 0.625, 0.628, 0.65, 0.66
59(b)	=	1	B1 for evaluating $4^{-2} = \frac{1}{16} = 0.0625$
60(a)	$\left(\frac{4}{9}\right)^{\frac{1}{2}}, 66\%, 0.65, \frac{5}{8}$	2	M1 for evaluating index term to $\frac{2}{3}$ and finding matching decimal benchmarks: 0.666..., 0.65, 0.66, 0.625
60(b)	=	1	B1 for evaluating $-\frac{7}{5} = -1.4$
61(a)	$67.5\%, \frac{17}{25}, 0.682, \left(\frac{1}{8}\right)^{-1} \times 10^{-1}$		M1 for evaluating individual terms to equivalent forms: 0.68, 0.675, 0.682, 0.8
61(b)	= or $\geq$ or $\leq$	1	B1 for verifying both sides are structurally equivalent to 0.035
62(a)	$-\frac{7}{8}$	2	M1 for finding decimal absolute baselines: 0.75, 0.80, 0.82, 0.875 to locate the largest distance from zero
62(b)	>	1	B1 for identifying $\left(\frac{27}{8}\right)^{-\frac{1}{3}} = \frac{2}{3} \approx 0.666... > 0.6$
63(a)	$-\frac{7}{8}$	2	M1 for finding decimal absolute baselines: 0.75, 0.80, 0.82, 0.875 to locate the largest distance from zero

Part	Answer	Marks	Partial Marks / Guidance
63(b)	>	1	B1 for identifying $\left(\frac{27}{8}\right)^{-\frac{1}{3}} = \frac{2}{3} \approx 0.666\ldots > 0.6$
64(a)	0.8485	1	B1
64(b)	C, B, A, D	2	B2 for all correct or B1 for 3 in correct relative order
64(c)	=	1	B1
65(a)	0.84	1	B1
65(b)	0.842	1	B1
65(c)	R, Q, P, S	2	B2 for all correct or B1 for 3 in correct relative order
66(a)	0.873	1	B1
66(b)	V_2, V_3, V_4, V_5, V_1	2	B2 for all correct or B1 for 3 or 4 in correct relative order
66(c)	>	1	B1
67(a)	0.6283	1	B1
67(b)	P, K, N, M, L	2	B2 for all correct or B1 for 3 or 4 in correct relative order
67(c)	>	1	B1
68(a)	1.433	1	B1
68(b)	S, Q, R, P	2	B2 for all correct or B1 for 3 in correct relative order
68(c)	>	1	B1
69(a)	8	1	B1
69(b)	Q, S, P, R	2	B2 for all correct or B1 for 3 in correct relative order
69(c)	<	1	B1

Part	Answer	Marks	Partial Marks / Guidance
70(a)	$\left(\frac{27}{8}\right)^{\frac{2}{3}} = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$	2	M1 for reciprocating fraction or taking cube root
70(b)	X_1, X_2, X_3, X_5, X_4	2	B2 for all correct or B1 for 3 or 4 in correct relative order
70(c)	$>$	1	B1
71(a)	-5	1	B1 for -5
71(b)	11	1	B1 for 11
72(a)	$\frac{7}{10}$	1	B1 for $\frac{7}{10}$ or equivalent fraction e.g. $\frac{6}{10} + \frac{1}{10}$
72(b)	$\frac{8}{15}$	1	B1 for $\frac{8}{15}$ or M1 for $\frac{4 \times 2}{1 \times 15}$ after cancellation
73(a)	$\frac{1}{2}$	2	M1 for conversion to improper fractions: $\frac{7}{3} - \frac{11}{6}$ or $\frac{14}{6} - \frac{11}{6}$
73(b)	$3\frac{1}{3}$	2	M1 for $\frac{5}{4} \times \frac{8}{3}$ or $\frac{40}{12}$ or equivalent mixed number fraction
74(a)	5.25	1	B1 for 5.25
74(b)	0.024	1	B1 for 0.024
75(a)	23	2	M1 for correct order of operations leading to $5 - 3 \times (-8) - 6$ or $5 + 24 - 6$
75(b)	$-\frac{11}{7}$	2	M1 for numerator processing to -11 or denominator processing to 7
76(a)	$1\frac{23}{28}$	2	M1 for evaluating the multiplication first: $\frac{5}{3} \times \frac{6}{7} = \frac{10}{7}$
76(b)	$2\frac{2}{5}$	2	M1 for brackets evaluating to $\frac{32}{10}$ or $\frac{16}{5}$ and M1 for multiplying by $\frac{3}{4}$
77(a)	-11	2	M1 for $3 - 8 = -5$ or $5 \times (-5)$
77(b)	8.5	2	M1 for $1.2 \div 0.3 = 4$
78(a)	-24	1	A1 for correct value

Part	Answer	Marks	Partial Marks / Guidance
78(b)	$2\frac{3}{8}$ or $\frac{19}{8}$	2	M1 for $\frac{7}{4} + \frac{5}{8} = \frac{14}{8} + \frac{5}{8}$
79(a)	6	2	M1 for $3.5 - 1.1 = 2.4$ or $\frac{2.4}{0.4}$
79(b)	$\frac{1}{9}$	2	M1 for cancellation or $\frac{20}{180}$
80(a)	$(3 + 5) \times 2 - 4 = 12$	1	A1 for correct bracket placement
80(b)	$\frac{3}{10}$	2	M1 for positive sign or $\frac{6}{20}$
81(a)	-3	2	M1 for $3 \times 4 = 12$ leading to $7 - 12 + 2$
81(b)	20	1	A1 for correct value
82(a)	3^6	1	B1
82(b)	5^3	1	B1
82(c)	2^{12}	1	B1
83(a)	1	1	B1
83(b)	$\frac{1}{16}$	1	B1
83(c)	$\frac{3}{8}$	1	B1
84(a)	5	1	B1
84(b)	3	1	B1
84(c)	8	2	B2 for correct answer, or M1 for $(\sqrt[4]{16})^3$ or 2^3 or $\sqrt[4]{4096}$
85(a)	2^2 or 4	1	B1
85(b)	6^3 or 216	1	B1
85(c)	$2\frac{1}{4}$	2	M1 for $\left(\frac{3}{2}\right)^2$ or $\frac{1}{\left(\frac{2}{3}\right)^2}$ or $\frac{2^{-2}}{3^{-2}}$ or $\frac{9}{4}$
86(a)	$\frac{1}{2}$	2	B2 for correct answer, or M1 for $\frac{1}{8^{\frac{1}{3}}}$ or $\frac{1}{2}$ seen in working

Part	Answer	Marks	Partial Marks / Guidance
86(b)	$\frac{2}{3}$	2	B2 for correct answer, or M1 for $\left(\frac{4}{9}\right)^{\frac{1}{2}}$ or $\frac{1}{\sqrt{\frac{9}{4}}}$ or $\frac{3}{2}$ seen
87(a)	4	2	M1 for $8^{\frac{2}{3}}$ or $(\sqrt[3]{8})^2$ or 2^2 or $\left(\frac{1}{4}\right)^{-1}$
87(b)	$\frac{5}{8}$	2	M1 for $125^{\frac{1}{3}} = 5$ or $2^{-3} = \frac{1}{8}$ or $\frac{5}{2^3}$
88(a)	-4	2	M1 for $3^x = 3^{-4}$ or $\frac{1}{81} = \frac{1}{3^4}$
88(b)	$\frac{7}{4}$	2	M1 for $(2^2)^{2x-1} = 2^5$ or $2(2x-1) = 5$ or $4x-2 = 5$
89(a)	12	3	M2 for $\frac{2^5 \times 2^3 \times 3^3}{3^2 \times 2^6}$ or M1 for breaking down bases into prime factors: $6^3 = 2^3 \times 3^3$ or $4^3 = 2^6$
89(b)	1	2	M1 for $\left(5^{\frac{1}{2}}\right)^4 = 5^2$ and $10^2 = 2^2 \times 5^2$ leading to $\frac{5^2 \times 2^2 \times 5^2}{2^2 \times 5^3}$ or $\frac{25 \times 100}{4 \times 125}$
90(a)	$\frac{27}{8}$	2	M1 for $\left(\frac{81}{16}\right)^{\frac{3}{4}}$ or $\left(\frac{3}{2}\right)^3$ or $\frac{2^{-3}}{3^{-3}}$
90(b)	$\frac{1}{17}$	3	M1 for $4^{\frac{3}{2}} = 8$, M1 for $27^{\frac{2}{3}} = 9$, then leading to $(8+9)^{-1}$
91(a)	$\frac{3}{5}$	2	M1 for converting to improper fraction $\left(\frac{25}{9}\right)^{-\frac{1}{2}}$ or finding $\sqrt{\frac{9}{25}}$
91(b)	4	2	M1 for writing all terms with base 3: $\frac{3^{-5} \times (3^2)^2}{(3^3)^{-1}}$ or $3^{-5+4-(-3)}$ or 3^2
92(a)	$x = 1, y = -2$	4	M1 for $x + 2y = -3$, M1 for $3x + y = 1$, M1 for attempting to solve the simultaneous linear system, A1 for both values correct.
93(a)	3×4^{-x}	2	M1 for $243^{\frac{x}{5}} = (3^5)^{\frac{x}{5}} = 3^x$ leading to $\frac{3^x \times 4^{-x}}{3^{x-1}}$ or $3^{x-(x-1)} \times 4^{-x}$
93(b)	0.0712	2	M1 for $3 \times 4^{-2.7}$ or $3 \times 0.0237...$ or 0.07119... seen

Part	Answer	Marks	Partial Marks / Guidance
94(a)	30 800	1	B1
94(b)	0.006 1	1	B1
95(a)	8×10^8	2	M1 for $2 \times 4 \times 10^{3+5}$ or 800,000,000
95(b)	3×10^{-4}	2	M1 for $3 \times 10^{2-6}$ or 0.0003
96(a)	1.5×10^3	2	M1 for 15×10^2 or 1500
97(a)	3×10^5	2	M1 for 30×10^4 or 300 000
97(b)	2.5×10^{-3}	2	M1 for 0.25×10^{-2} or $\frac{1}{400}$ or 0.0025
98(a)	3.6×10^5	2	M1 for $43 \times 10^4 - 7 \times 10^4$ or 430 000 – 70 000 or 360 000
99(a)	–3	2	M1 for 0.009 or $(4.5 \times 10^{-5}) \times (2 \times 10^2) = 9 \times 10^{-3}$
99(b)	–6	2	M1 for $\frac{4}{1\,000\,000}$ or 0.000004
100(a)	2.98×10^{-3}	2	M1 for $24 \times 10^{-4} + 5.8 \times 10^{-4}$ or 0.0024 + 0.00058
100(b)	1.44×10^8	2	M1 for 1.44×10^k or 144 000 000
101(a)	7×10^2	3	M1 for numerator simplified to 14×10^4 or 1.4×10^5 , M1 for division step $\frac{1.4 \times 10^5}{2 \times 10^2}$ or $\frac{140\,000}{200}$
102(a)	4.5×10^n	2	M1 for factorising out 10^{n-1} to get $(4 \times 10 + 5) \times 10^{n-1} = 45 \times 10^{n-1}$ or for equating powers $4 \times 10^n + 0.5 \times 10^n$
102(b)	$3.2 \times 10^{n+1}$	2	M1 for $\frac{16 \times 10^{2n}}{5 \times 10^{n-1}}$ or for $3.2 \times 10^{2n-(n-1)}$
103(a)	5×10^4	3	M1 for converting numerator to same power: $12 \times 10^{-4} + 8 \times 10^{-4} = 20 \times 10^{-4}$ or 2×10^{-3} , M1 for dividing by 4×10^{-8} to get 0.5×10^5
104(a)	–1	3	M1 for $(6.25 \times 10^{2k}) \times (8 \times 10^3)$, M1 for $50 \times 10^{2k+3} = 5 \times 10^{2k+4}$, A1 for equating $2k+4 = 2$ to find $k = -1$

Part	Answer	Marks	Partial Marks / Guidance
104(b)	8×10^{-15}	2	M1 for $\frac{1}{125 \times 10^{12}}$ or 0.008×10^{-12}
105(a)	2.1×10^{-5}	2	M1 for $30 \times 10^{-6} - 9 \times 10^{-6} = 21 \times 10^{-6}$ or $3 \times 10^{-5} - 0.9 \times 10^{-5}$
105(b)	1×10^{-2}	3	M1 for $X^2 = 9 \times 10^{-10}$, M1 for $\frac{9 \times 10^{-10}}{9 \times 10^{-6}} = 1 \times 10^{-4}$ or 10^{-6} error in division, A1 for $\sqrt[3]{10^{-6}} = 10^{-2}$
106(a)	2.5×10^{-1}	2	M1 for numerator = 25×10^{-6} and denominator = 1×10^{-4} leading to 25×10^{-2}
106(b)	2.44×10^{-4}	3	M1 for $\frac{1}{M} = 4100$ or 4.1×10^3 , M1 for $M = \frac{1}{4100} = \frac{1}{4.1} \times 10^{-3}$ or showing step leading to $0.0002439...$ or $\frac{10}{41} \times 10^{-4}$
107(a)	4.39×10^7	2	M1 for $42 \times 10^6 + 1.9 \times 10^6$ or $4.2 \times 10^7 + 0.19 \times 10^7$ or 43 900 000
107(b)	8.78×10^3	2	M1 for $\frac{4.39 \times 10^7}{5 \times 10^3}$ or 8780
108(a)	5.73×10^8	2	M1 for $5.82 \times 10^8 - 0.094 \times 10^8$ or $582 \times 10^6 - 9.4 \times 10^6$ or 572 600 000
108(b)	1.62%	2	M1 for $\frac{9.4 \times 10^6}{5.82 \times 10^8} \times 100$ or $0.016151... \times 100$
109(a)	30	2	M1 for $\frac{7.5 \times 10^{11}}{2.5 \times 10^{10}}$ or $\frac{75 \times 10^{10}}{2.5 \times 10^{10}}$
109(b)	3.88×10^{11}	2	M1 for $\frac{7.5 \times 10^{11} + 0.25 \times 10^{11}}{2}$ or $\frac{7.75 \times 10^{11}}{2}$ or 387 500 000 000
110(a)	4.85×10^{14}	2	M1 for $4.76 \times 10^{14} + 0.0895 \times 10^{14}$ or 484.95×10^{12} or 484 950 000 000 000
110(b)	53.2	2	M1 for $\frac{4.76 \times 10^{14}}{8.95 \times 10^{12}}$ or 53.1843...
111(a)	6.23×10^2	3	M1 for multiplying numerator to 7.7872×10^4 , M1 for dividing by 1.25×10^2 to get 622.976, A1 for standard form to 3 sig figs
111(b)	623	1	B1 ft from part (a)
112(a)	483.71	1	B1

Part	Answer	Marks	Partial Marks / Guidance
112(b)	484	1	B1
113(a)	1000	3	M1 for writing individual values to 1 sig fig as 40, 5, and 0.2, M1 for $\frac{40 \times 5}{0.2}$ or $\frac{200}{0.2}$, A1 for 1000.
114(a)	0.0030	1	B1
114(b)	80 000	1	B1
115(a)	10	2	M1 for $\frac{40 \times 0.05}{\sqrt{0.04}}$ or $\frac{2}{0.2}$
115(b)	0.05	1	B1
116(a)	0.06	2	M1 for showing individual 1 sig fig roundings: $\frac{6 \times 0.2}{200}$ or $\frac{1.2}{200}$ or $\frac{12}{2000}$
116(b)	$\frac{3}{50}$	1	B1
117(a)	£70	2	M1 for $\frac{140}{2+3+5} \times 5$ oe
117(b)	£42	2	M1 for $(5 - 2) \times 14$ or $70 - 28$ oe
118(a)	300 g	2	M1 for $\frac{120}{8} \times 20$ or 15×20
118(b)	30	2	M1 for $\frac{450}{15}$ oe
119(a)	8 kg	2	M1 for $\frac{14}{7} \times 4$ oe
119(b)	42 kg	2	M1 for $3.5 \times (7 + 4 + 1)$ oe
120(a)	600 ml	2	M1 for $\frac{3}{5+3+1} \times 1800 = \frac{3}{9} \times 1800$
120(b)	5 : 3 : 2	2	M1 for doubling the last component to get 5 : 3 : 2 visible or structured
121(a)	$\frac{3}{10}$	1	Correct answer only
121(b)	60 kg	3	M1 for $7 - 3 = 4$ parts, M1 for 1 part = $\frac{24}{4} = 6$ kg, A1 for $10 \times 6 = 60$ kg
122(a)	20 cm	2	M1 for $\frac{3000}{150}$ oe
122(b)	900 m ²	3	M1 for 400×150^2 , M1 for division by 10000 to convert cm ² to m ²

Part	Answer	Marks	Partial Marks / Guidance
123(a)	3 : 8	2	M1 for 45 : 120 or correct conversion of units to common form
124(a)	$\frac{7}{9}$	1	Correct answer only
124(b)	Proof	2	M1 for expressing salt as $\frac{2}{9}V$ and new water as $\frac{7}{9}V + 60$, M1 for equating new water to $4 \times \text{salt} = 4 \left(\frac{2}{9}V \right) = \frac{8}{9}V$
124(c)	540 ml	2	M1 for isolating $\frac{1}{9}V = 60$ or equivalent shift
125(a)	$n = 200$	3	M1 for actual area = $12 \times 16 = 192 \text{ m}^2$, M1 for converting to cm^2 : $192 \times 10000 = 1920000 \text{ cm}^2$, A1 for $n^2 = \frac{1920000}{48} = 40000 \Rightarrow n = 200$
125(b)	27 cm	2	M1 for $\frac{5400}{200}$ oe
126(a)	£1200	2	M1 for $\frac{4}{12} \times 3600$ oe
126(b)	Manager C, loss of £100	3	M1 for finding new shares: $A = 800, B = 1200, C = 1600$, M1 for comparing old shares ($A = 900, B = 1200, C = 1500$) to identify A loses £100 and C gains £100, A1 for Manager A with £100 loss
127(a)	3 hours 30 minutes	2	M1 for $\frac{1400}{400}$ or 3.5 hours
127(b)	600	2	M1 for $\frac{1400}{2\frac{1}{3}}$ or $\frac{1400 \times 3}{7}$
128(a)	5.4	3	M1 for $15 \times 60 = 900$ seconds, M1 for $6 \times 900 = 5400 \text{ m}$
128(b)	$\frac{6 \times 3600}{1000} = 21.6$	2	M1 for 6×3600 or $\frac{6}{1000}$ seen
129(a)	12	2	M1 for $\frac{45}{3}$ or $\frac{45 \times 4}{15}$ $3\frac{3}{4}$
129(b)	2.7	2	M1 for $45 \times \frac{6}{100}$ or $\frac{270}{100}$

Part	Answer	Marks	Partial Marks / Guidance
130(a)	$18\frac{4}{7}$	3	M1 for total distance = 36, M1 for total time = $\frac{12}{16} + \frac{24}{20} = \frac{3}{4} + \frac{6}{5} = \frac{39}{20}$ hours, A1 for $\frac{36 \times 20}{39} = \frac{240}{13}$ or $18\frac{6}{13}$
130(b)	4.6	2	M1 for mass = $4000 \times 1.15 = 4600$ g
131(a)	265	2	M1 for distance of second part = $100 \times 1\frac{3}{4} = 175$ km
131(b)	106	2	M1 for total time = $\frac{90}{120} + 1\frac{3}{4} = \frac{3}{4} + \frac{7}{4} = 2.5$ hours, then $\frac{265}{2.5}$
132(a)	$\frac{2x}{\frac{x}{60} + \frac{x}{90}} = 72$	3	M1 for setting total distance as $2x$, M1 for total time expression $\frac{x}{60} + \frac{x}{90} = \frac{5x}{180} = \frac{x}{36}$, A1 for $\frac{2x}{\frac{x}{36}} = 72$
132(b)	120	2	M1 for total distance = $72 \times 3\frac{1}{3} = 72 \times \frac{10}{3} = 240$ km, then $\frac{240}{2}$
133(a)	1620	1	B1 for 1620
133(b)	751.68	2	M1 for $\frac{2}{5} \times 1620 = 648$, then 648×1.16
133(c)	29	2	M1 for fuel volume = $\frac{250}{100} \times 8 = 20$ litres, then 20×1.45
134(a)	$\frac{2v(v-15)}{(v-15)+v} = \frac{2v^2-30v}{2v-15}$	3	M1 for total distance = $2d$ and time expressions $\frac{d}{v}$ and $\frac{d}{v-15}$
134			M1 for adding times to get $\frac{d(v-15)+dv}{v(v-15)} = \frac{d(2v-15)}{v(v-15)}$
134(b)	60	3	M1 for $\frac{2v^2-30v}{2v-15} = 40 \Rightarrow 2v^2-30v = 80v-600$
134			M1 for $2v^2-110v+600=0 \Rightarrow (2v-20)(v-60)=0$

Part	Answer	Marks	Partial Marks / Guidance
135(a)	$\frac{\frac{d}{\left(\frac{d}{3}\right)}}{x} + \frac{\frac{\left(\frac{2d}{3}\right)}{\left(\frac{2d}{3}\right)}}{2x} = \frac{3}{2}x$	3	M1 for total time = $\frac{d}{3x} + \frac{2d}{6x}$
135			M1 for simplifying denominator to $\frac{2d}{3x}$
135			A1 for $\frac{d}{\left(\frac{2d}{3x}\right)} = \frac{3x}{2}$
135(b)	48	2	M1 for equating $\frac{3}{2}x = \frac{270}{3\frac{3}{4}} = \frac{270 \times 4}{15} = 72$
136(a)	1 hour 30 minutes	2	M1 for $\frac{114}{76}$
136(a)(i)	74.7 km/h	3	M1 for total time = $1.5 + 2.3 = 3.8$ hours
136(a)(ii)			M1 for $\frac{284}{3.8}$
137(a)	EUR 522.90	2	M1 for 450×1.162
137(a)(i)	62.0 m/s	3	M1 for total seconds = $82 \times 60 = 4920$ s
137(a)(ii)			M1 for $\frac{305000}{4920}$
138(a)	CAD 6753	3	M1 for $3850 \times 1.284 = 4943.4$ USD
138			M1 for $\frac{4943.4}{0.732}$
138			A1 for 6753.28 rounded to 6753
138(b)	GBP 89.10	4	M1 for fuel volume = $\frac{645}{100} \times 14.8 = 95.46$ litres
138			M1 for cost in CAD = $95.46 \times 1.64 = 156.5544$ CAD
138			M1 for converting CAD to GBP: $\frac{156.5544 \times 0.732}{1.284}$
138			A1 for 89.10 (allow 89.09 to 89.11)
139(a)	280 GBP	2	M1 for $250 \times \frac{112}{100}$ or $250 + 30$

Part	Answer	Marks	Partial Marks / Guidance
139(b)	252 GBP	2	M1 for their $280 \times \frac{90}{100}$ or their $280 - 28$
140(a)	150 EUR	3	M2 for $180 \div \frac{120}{100}$ or $180 \times \frac{100}{120}$ or M1 for identifying $120\% = 180$
141(a)	605 grams	3	M2 for $500 \times \left(\frac{110}{100}\right)^2$ or $500 \times 1.1 \times 1.1$ or M1 for $500 \times 1.1 = 550$
141(b)	121%	1	B1 for 121% or their $\frac{605}{500} \times 100$
142(a)	1000 GBP	2	M1 for $800 \times \left(1 + \frac{25}{100}\right)$ or $800 \times \frac{5}{4}$
142(a)(i)	6.25%	3	M1 for their $1000 \times \frac{85}{100}$ or 850
142(a)(ii)			M1 for $\frac{\text{their } 850 - 800}{800} \times 100$ or $\frac{5}{4} \times \frac{85}{100} = \frac{425}{400} = 1.0625$
143(a)	2646 EUR	2	M1 for Year 1: $2400 \times 1.05 = 2520$
143(a)(i)			M1 for Year 2: their $2520 \times 1.05 = 2646$ with no dots on line
143(a)(ii)	132.30 EUR	2	M1 for $2778.30 - 2646$ or $2646 \times \frac{5}{100}$
144(a)	$\left(1 + \frac{a}{100}\right) \left(1 - \frac{a}{100}\right) = \frac{24}{25} \Rightarrow 1 - \left(\frac{a}{100}\right)^2 = \frac{24}{25}$	3	M1 for product of multipliers or $P \left(1 - \frac{a^2}{10000}\right)$
144(a)(i)	$\Rightarrow \left(\frac{a}{100}\right)^2 = 1 - \frac{24}{25} = \frac{1}{25}$		M1 for clear algebraic simplification to given form
144(a)(ii)	20	1	B1 for 20 (reject -20)
144(a)(iii)	50000 GBP	2	M1 for $48000 \div \frac{24}{25}$ or $48000 \times \frac{25}{24}$
145(a)	1039.50 GBP	2	M1 for $4500 \times 0.0385 \times 6$
145(b)	5539.50 GBP	1	FT for $4500 + \text{their (a)}$
146(a)	141000 USD	3	M2 for $125000 \times (1.0425)^3$ or M1 for 125000×1.0425
146(b)	16400 USD	1	FT for their (a) $- 125000$
147(a)	18742.66 EUR	2	M1 for 14500×1.0435^6

Part	Answer	Marks	Partial Marks / Guidance
147(b)	18030.44 EUR	2	M1 for Ans from (a) $\times (1 - 0.038)$
147(c)	124.3% or 124%	2	M1 for $\frac{\text{Ans from (b)}}{14500} \times 100$
148(a)	196620 AUD	3 M1 for $480000 \times 0.0565 \times 7.25$	M1 for $t = 7.25$ 148
148(b)	Show derived statement	1	A1 for explicit statement showing $1 + \frac{4.95}{100} = 1.0495$ applied compoundingly
148(c)	11 years	2 A1 for $n = 11$ (since 10 years gives ≈ 599650)	M1 for systematic trial or setting up equation $370000 \times 1.0495^n > 600000$ 148
149(a)	06 15	2	M1 for bridging midnight, e.g., 22 45 + 1 hour 15 minutes is 00 00, leaving 6 hours 15 minutes.
149(b)	Wednesday	1	B1 for the correct day following midnight.
150(a)	18 15	1	B1 for correct addition of 4 hours.
150(b)	110 minutes	2	M1 for 1 hour 50 minutes or $60 + 50$.
151(a)	13 20	1	B1 for subtracting 2 hours from departure time.
151(b)	18 20 on 13 April	2	M1 for adding 3 days to the date (13 April) and 5 hours to the time ($1320 + 5$ hours).
152(a)	10:45 p.m.	1	B1 for correct time format with p.m.
152(b)(i)	19 20	2	M1 for adding 9 hours and 11 hours 35 minutes to 22 45 or equivalent

Part	Answer	Marks	Partial Marks / Guidance
152(b)(ii)	Wednesday	1	B1 for Wednesday
153(a)	Tuesday 20 20	2	M1 for correct time zone shift to departure local time ($2145 + 9 \text{ hours} = \text{Tuesday } 0645$) or adding duration first.
153(b)	13.6	2	M1 for $\frac{35}{60}$ seen or 13.583...